Tecnológico de Monterrey

Teaching Notes
Molson Coors Brewing Co.: Valuing a Convertible Bond

NE90-06-107

Synopsis

In 2007, Deutsche Bank became the lead book runner by issuing $575 million convertible bonds for Molson Coors Brewing Co. Being the first convertible bond issuance for Molson Coors, required a considerable amount of effort from Deutsche's Equity Derivatives Group to convince Molson Coors management to choose them over its competitors. This case focuses on Deutsche's choice of the conversion premium and coupon rate to be proposed to Molson Coors management. This pricing decision requires students to understand the concept of valuing a convertible as the sum of a straight bond plus the conversion option. Valuing the conversion option as a call option requires the use of the Black-Scholes model.

Objectives

The case is designed for students who already have a basic knowledge of bond valuation and option-pricing principles. Because the case touches on both technical and strategic issues, it works well with undergraduate and MBA students. The instructor may choose to hand out **Exhibit 1** at the end of the class as it serves as the *re-cap* or this business case.

Suggested Questions for Advance Assignment

1. What are Molson Coors uses of funds for 2007? How important is the tender offer (bond repurchase) to Molson Coors choice to issue a convertible bond?
2. What are the pros and cons of issuing convertible debt vs. straight debt or equity?
3. How can you explain that a convertible can be valued as the sum of a straight bond plus a call option?
4. As Molson Coors CEO, what do you like and not like about this proposal from Deutsche Bank? In particular, do you like the 25% conversion premium and the coupon rate?

This teaching notes were prepared by Professor Roberto Seira Rojas (robertoseira@tec.mx) for the sole purpose of aiding instructors in the use of the case "Molson Coors Brewing Co.: Valuing a Convertible Bond". The teaching notes provide analysis and questions about the case, that are intended to present alternative approaches to help the student's comprehension of the topics discussed.

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Centro Internacional de Casos
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5. How can we change Molson Coors financially engineered straight bond into a convertible bond with a 60% conversion premium?

6. Do the following calculations:

a. Calculate Coupon Rate for the convertible that would result in the debt being issued at exactly the face value of $1,000 per bond.
b. If Convertible Bond Value = Bond Value + Conversion Option Value, calculate:
    a. Bond Value
    b. Conversion Option Value: use Black-Scholes model (European Option / Basic / No Dividend / Model).

For Question 6, use the following assumptions:
a. Risk free rate = 4.46% (5yr Treasury rate)
b. Annualized volatility of Molson Coors stock = 23%
c. Years to expiration = 6
d. Discount rate = 5.75%

7. How much should Molson Coors pay to buy the call options? How much should Molson Coors realize from the sale of warrants?

Questions and answers

1. What are Molson Coors uses of funds for 2007? How important is the tender offer (bond repurchase) to Molson Coors choice to issue a convertible bond?

   This question is designed to focus the class on the bond repurchase program as a critical use of funds for 2007. The primary advantage of bond repurchases is the interest expense reduction for the company. Students should notice that most of the $575 million raised will be used to repurchase bonds ($465 million).

2. What are the pros and cons of issuing convertible debt vs. traight debt or equity?

   a) From the investor's point of view:

   Pros:
   - With convertibles, the investor has unlimited upside potential of the stock, with protected down side with the bond.
   - In case of conversion, the investor turns into a shareholder of the company.

   Cons:
   - A convertible bond has a lower coupon than and identical straight bond.

   b) From the company's point of view:

   Pros:
   - Convertible debt allows the firm to sell equity at a higher price than at the time of the convertible bond issuance.
   - Debt is tax deductible, which makes this financing strategy even cheaper.
   - The company has not to pay any dividends until conversion of the bond.
   - If the company does well, it is better to issue a convertible bond rather than equity.

   Cons:
   - If converted, there will be a dilutive effect on the shares outstanding.

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3. The case states a convertible bond can be valued as the sum of a straight bond plus a call option.

a) Starting with the current stock price of $87.61 per share, how can you use the Black-Scholes model to estimate the value of the conversion option with a 25% conversion premium for one share of Molson Coors stock?
Much of the class time will be devoted to the technical aspect of pricing the convertible and, in particular to choosing the correct input values for the Black-Scholes pricing model to estimate the value of the conversion option. A key conceptual point will be valuing the conversion option at the bond level, rather than at the share level. Many students, however, will have difficulty translating the per-share call option value to the value of the conversion option for a $1,000 bond. Students will eventually recognize that the conversion price can be used to find the number of shares per bond received upon conversion. **A key lesson of the case is the ability to use the model at the correct scale; i.e., either to value a bundle of options within a bond or to value a single option**. **Exhibit 2** shows the option calculations both on a per-share basis and on a per-bond basis.

b) What is the value of the straight bond component? What coupon rate should Deutsche Bank propose in order for the convert to sell at exactly $1,000 per bond?
This question relates to the second step in valuing the convertible: solving for the coupon rate on the straight bond component such that the total value of the convertible sums to $1,000 per bond. A key point of emphasis will be the discount rate used to value the bond cash flows. Students may be surprised at how low the coupon rate is, relative to the discount rate (YTM) of 5.75%.

Conceptually, what should happen to the coupon rate if Deutsche were to propose a 15% and a 40% conversion premium? Students should see that the lower conversion premium is synonymous with the option being closer to the money and therefore carrying a higher value. As the option component is increased, the bond component must be decreased by lowering the coupon rate. This illustrates why convertibles have lower coupon rates than straight debt and why the convertible coupon could be extremely low if the conversion premium were also very low.

The straight debt valuation is presented in **Exhibit 3**. Since the maturity (six years) and face value ($1,000) are known, we only need the coupon rate and appropriate discount rate to estimate the value of the straight debt. Deutsche's task is to set the coupon rate such that the convertible will sell for full value; which is to say, $1,000 per bond. With the conversion premium estimated as $191.5, the coupon should be set so that the bond value equals the remaining value of $808.5. In fact, as shown in **Exhibit 3**, a 1.93% coupon creates a debt value of $808.5, which when added to the conversion option value gives $1,000.

4. As Molson Coors CEO, what do you like and not like about this proposal from Deutsche Bank? In particular, do you like the 25% conversion premium and the coupon rate?

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Cons:
As Molson Coors CEO we would think that the 25% conversion premium is too small, taking into consideration that the company has performed very well in the past and future growth is expected.

The stock price will probably reach the exercise price in a very short period of time, and that is a risk that needs to be considered because it might mean that investors would convert the bond sooner than expected. This might not be necessarily good for the company by not giving it enough time to fully take advantage of the cheaper cost of financing.

Pros:
As previously discussed, the coupon rate on a convertible bond is lower than the rate of a straight bond. This proposal implies a smart and cheaper financing strategy for the company. At the same time, it is also attractive to the investors because it offers them unlimited upside potential with a limited downside.

5. How can we change Molson Coors financially engineered straight bond into a convertible bond with a 60% conversion premium?
   A good way to begin this analysis is to remind the students that a convertible can be valued as the sum of a straight bond plus a warrant/option. Therefore, **for each convertible bond issued by Molson Coors, the company is short a bond and short a bundle of call options. The short options can be nullified, if Molson Coors buys call options with the same terms (exercise price and maturity)**. The combination of the convertible plus the call options is equivalent to having issued a straight bond. Finally, a new synthetic convertible can be created by Molson Coors selling warrants with a strike price set at a 60% conversion premium; which is to say, 60% higher than the stock price at time of issuance.

6. How much should Molson Coors pay to buy the call options? How much should Molson Coors realize from the sale of the warrants?
   Understanding the concepts behind the hedge is important, but the pricing is the key to seeing that there is no free lunch for Molson Coors. No matter how it is done, Molson Coors ultimately must pay for the higher conversion premium. Using the Black-Scholes model, the class will see that the cost of buying the call options is higher than the value received from selling the warrants (assuming the same number of equivalent shares). See **Exhibit 4.**

7. Epilogue.
   The instructor should end the class by distributing copies of **Exhibit 1**. Discussion points will center on the actual terms of Molson Coors deal with particular emphasis upon the low coupon rate. The instructor can conclude the class with a discussion about how Deutsche was able to alter the effective conversion premium by having Molson Coors buy and sell warrants with differing strike prices.

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Exhibit 1: Convertible Issuance Re-Cap

- Issued $500M 6 years convertible debentures due July 2013 on June 11, 2007
  - — 2.5% coupon, 25% conversion premium = $109.5 stock price (based on closing stock price day of $87.61)
  - — Raised conversion price to 60% premium ($140.2 stock price) with a call spread overlay at very competitive terms
    - Private transaction with Deutsche Bank, Morgan Stanley and Citibank
  - — Bookrunners (Deutsche Bank and Citibank); Co-managers (JP Morgan, Morgan Stanley, Bank of Montreal and Wachovia)
- Upsized to $575M due to very strong demand
  - — Deal was 4x oversubscribed. Coupon and conversion premium priced in the mid-range despite adverse treasury rate movements
- Use of funds

$465M to buy back a portion of $850M 6.38% 2012 bonds
$50M additional US pension contribution
$60M cost of call spread and fees
$575M Total

I. Convertible (a)

Ranking	Senior Unsecured
Proceeds	$575 million
Maturity	6
Call protection	Non Call - Life
Investor puts	None
Coupon / yield-to-maturit	2.5%
Conversion premium	25%
Conversion price(a)	$109.51

II. Note hedge and call warrant

Convertible note hedge strike premi	25%	
Call warrant strike premium	60%	
	% of principal	Per $575m
Cost of convertible note hedge	(18.6%)	($106.7)
Proceeds from call warrant	9.9%	$57.0
Total	(8.6%)	($49.7)
Combined transaction		
Convertible underlying shares(b)	5.3 million	
Effective conversion premium	60%	
Effective conversion price	$140.18	

(a) Based on closing stock price on pricing day of $87.61.
(b) Based on closing stock price at issuance of $87.61. Equal number of shares underlying convert, note hedge and call warrant.

Note: Conversion Ratio = Par Value/Conversion Price =$575M/$109.51=5.3M shares.

Debt Tender Re-Cap

- Announced a partial tender to repurchase $450M of the $850M notional 6.38% 2012 bonds concurrently with the convertible announcement
  - — Offered investors a spread at dollar price of $103.3, which was $1.1 above the trading price of $102.2, to achieve our target tender size
  - — JP Morgan acted as sole dealer-manager

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- Upsized the tender to $625mm due to very strong demand, allowing Molson Coors to buy back a larger amount without increasing the price paid to investors
- Entered into a 'T-lock', to mitigate some risk from Treasury rate movement, that settled for +$1.4M
- Cash outlay by TAP for the tender of $645M, including premium of $20M above par [625,000 pieces of $1000* ($1,033 -$1,000 = premium over par value)]
    - Given the premium and associated dealer fees, TAP will incur a one-time, upfront accounting charge to interest expense.

Rationale For The Transaction: Boost Profit

Lower interest and pension expenses, improve profit trends.

A. Debt Repurchase
    1. Replace high-coupon 6-3/8% debt with much lower coupon convertible
    2. Reduce total company pretax interest expense approximately $15M annually
    3. Tax benefits of convertible option hedge
    4. Largely balance-sheet neutral: existing debt replaced with lower-cost funds
B. Pension Contributions
    1. Fund US pension deficit with low-cost funds, which reduces pension expense
    2. Accelerate lowering of pension exposure via investment of contributions in fixed income
    3. Capture tax benefits of pension contributions to further reduce cost of funds
    4. Balance sheet neutral as pension obligations replaced one-for-one with lower-cost debt

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Combined Sources and Uses of Proceeds / Cash

Total Convertible Issuance	$575.0
Underwriter fees (1.625%)	($9.3)
Net Convertible Proceeds	$565.7
Total Note Hedge Cost	***($106.7)***
Total Warrant Proceeds	***$57.0***
Net Call Spread	**($49.7)**
Net Convertible and Call Spread Proceeds	$516.0
US Pension Contribution	($50.0)
Amount available for Debt Repurchase / Costs	$466.0
Fees / Costs	
Debt Tender Premium	($20.7)
Dealer fees (For Debt Tender)	($1.5)
Convertible and Debt Tender Related Fees	($1.0)
Available for Debt Repurchase after Costs	$442.8
In-the-money Swap (Reverse T-lock)	*$1.4*
Accrued Interest	*($6.2)*
Amount funded with Cash	$187.0

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Exhibit 2

MOLSON COORS
Conversion Option Value

Current stock price	$87.61	
Conversion premium	25%	
Conversion price	$109.51	
Conversion shares	9.13	
Conversion option value	**$191.5**	**Call Value**

Black-Scholes Inputs per Share
(no dividend, European option)

S: underlying asset price	$87.61	**Call Value**	**$21.0**
X: exercise price	$109.51	Call Delta (hedge ratio)	0.64
R_f: risk-free rate	4.46%		
σ: annualized volatility	23.0%	Using Put-Call Parity	
t: years to expiration	6.00	Put Value	$17.2

Conversion option value = call value per share x shares per bond = **$191.5**

Black-Scholes Inputs per Bond
(no dividend, European option)

S: underlying asset price	$800	**Call Value**	**$191.5**
X: exercise price	$1000	Call Delta (hedge ratio)	0.64
R_f: risk-free rate	4.46%		
σ: annualized volatility	23.0%	Using Put-Call Parity	
t: years to expiration	6.00	Put Value	$156.7

Delta, also called the hedge ratio, measures the sensitivity of the price of an option to the price of its underlying asset. For example, a delta of 0.3 for a call option indicates that the price of the call will rise $0.30 for each dollar increase in the price of the underlying asset. This price sensitivity also tell us how to create a hedge position by indicating how much of the underlying asset the option hedges. A delta of 0.5 for a call on common stock tell us that one call hedges 0.5 shares of stock. To hedge this position, an investor would write two calls for each share of stock held.

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Exhibit 3

MOLSON COORS
Straight Debt Value and Convertible Bond Value

Face Value	$1,000
Coupon Rate	1.93%
Discount Rate	5.75%
Years to maturity	6.0

Years =>	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
Coupon Payments	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65
Face Value	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$0	$1,000
Total Cash Flows	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$9.65	$1,009.65

Market Value: Coupons	$96.8
Market Value: Face Value	$711.7
Market Bond Value: Total	**$808.5**
Conversion option	**191.5**
Convertible Value	**$1,000** (Conversion option + Straight debt value)

n	12
i	2.875
PMT	9.65
FV	1000
PV	-808.5

Exhibit 4

MOLSON COORS
Valuing the Hedging Transaction

Purchase call options to nullify conversion option:

Call Options for 2013 Notes:

S: underlying asset price	$87.61	Call Value	$21.0
X: exercise price	$109.51	Call Delta (hedge ratio)	0.63
Conversion premium = 1000/667 =	25.0%		
R_f: risk-free rate	4.46%		
σ: annualized volatility	23.00%	Using Put-Call Parity	
t: years to expiration	6.00	Put Value	$17.2
		Put Delta (hedge ratio)	-0.37

$21 per option/share x 5.3 million shares = $110 ($millions)
Cost of Note Hedge

Estimate using "Conv. Bond" = $191.5* 5.75% $110 ($millions)
=Conv Opt Value * Disc Rate

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MOLSON COORS
Valuing the Warrant Issuance

Sell warrants with 60% conversion premium:

2013 Warrants: 6 years, conversion price = $109.51

S: underlying asset price	$87.61	Call Value	$11.4
X: exercise price	$140.2	Call Delta (hedge ratio)	0.47
Conversion premium	60.0%		
R_f: risk-free rate	4.46%		
σ: annualized volatility	21.00%	Using Put-Call Parity	
t: years to expiration	6.00	Put Value	$32.8
		Put Delta (hedge ratio)	-0.53

$11.4 per option/share x 5.3 million shares = $59 ($millions)
Proceeds from Call Warrants

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